EXPERIMENT :03

Write a C program for Playfair algorithm is based on the use of a 5 X 5 matrix of letters constructed using a keyword. Plaintext is encrypted two letters at a time using this matrix.

CODE:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
char matrix[5][5];
void createMatrix(char key[])
{
   char used[26] = {0};
   int r = 0, c = 0, i;
   used['J' - 'A'] = 1;
   for(i = 0; key[i] != '\0'; i++)
   {
      char ch = toupper(key[i]);
      if(ch == 'J')
         ch = 'I';
      if(ch >= 'A' && ch <= 'Z' && !used[ch - 'A'])
      {
         matrix[r][c++] = ch;
         used[ch - 'A'] = 1;
         if(c == 5)
         {
            c = 0;
            r++;
         }
      }
   }
```

```
    for(i = 0; i < 26; i++)
    {
        if(!used[i])
        {
            matrix[r][c++] = 'A' + i;
            used[i] = 1;
            if(c == 5)
            {
                c = 0;
                r++;
            }
        }
    }
}
void findPosition(char ch, int *row, int *col)
{
    int i, j;
    if(ch == 'J')
        ch = 'I';
    for(i = 0; i < 5; i++)
    {
        for(j = 0; j < 5; j++)
        {
            if(matrix[i][j] == ch)
            {
                *row = i;
                *col = j;
                return;
            }
        }
    }
```

```
}
int main()
{
    char key[50], text[200], plain[200];
    int i, j = 0;
    int r1, c1, r2, c2;
    printf("Enter the keyword: ");
    scanf("%s", key);
    createMatrix(key);
    printf("\nPlayfair Matrix:\n");
    for(i = 0; i < 5; i++)
    {
        for(j = 0; j < 5; j++)
            printf("%c ", matrix[i][j]);
        printf("\n");
    }
    printf("\nEnter the plaintext: ");
    scanf("%s", text);
    for(i = 0; text[i] != '\0'; i++)
    {
        char ch = toupper(text[i]);
        if(ch == 'J')
            ch = 'I';
        plain[j++] = ch;
    }
    plain[j] = '\0';
    printf("\nEncrypted text: ");
    for(i = 0; i < j; i += 2)
    {
        char a = plain[i];
        char b;
```

```
    if(i + 1 < j)
        b = plain[i + 1];
    else
        b = 'X';
    if(a == b)
    {
        b = 'X';
        i--;
    }
    findPosition(a, &r1, &c1);
    findPosition(b, &r2, &c2);
    if(r1 == r2)
    {
        printf("%c%c",
                matrix[r1][(c1 + 1) % 5],
                matrix[r2][(c2 + 1) % 5]);
    }
    else if(c1 == c2)
    {
        printf("%c%c",
                matrix[(r1 + 1) % 5][c1],
                matrix[(r2 + 1) % 5][c2]);
    }
    else
    {
        printf("%c%c",
                matrix[r1][c2],
                matrix[r2][c1]);
    }
}
```

```
    return 0;
}
```

OUTPUT:

